Supervised Learning

Complete Study Guide

Beginner to Intermediate Level

Subject	Chapter	Level	Sections
Machine Learning	1 – Supervised Learning	Beginner → Intermediate	9 Comprehensive Sections

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■ Section 1: Conceptual Notes

Core concepts explained in plain language with analogies and examples

What is Supervised Learning?

Supervised learning is a type of machine learning where a computer learns to make predictions or decisions by **studying labeled examples**. Think of it like a student learning with an answer key — for every practice question (input), the student can see the correct answer (output). Over many examples, the student figures out the pattern and can answer **new questions** it has never seen before.

> ■ *Key Analogy: A child learning to distinguish cats from dogs by being shown hundreds of labeled photos ('this is a cat', 'this is a dog') is performing supervised learning!*

Labeled Data: Features and Labels

Every training example in supervised learning has two parts: **features** (the inputs) and a **label** (the correct output/answer).

Concept	Also Called	Example	Real-world Meaning
Feature	Input / Attribute / X	House size, No. of rooms, Location	Measurable properties of the data
Label	Target / Output / Y	House price ($250,000)	The answer we want the model to predict
Training Set	Labeled Dataset	1,000 house records with prices	Collection of (feature, label) pairs for learning

Classification vs. Regression

Supervised learning problems fall into two main categories based on the type of output they produce:

■■ Classification	■ Regression

Predicts a **discrete category / class**. • Output: finite set of labels • Example: Is this email spam? → **Yes / No** • Example: What digit is this? → **0–9** • Example: Is the tumour malignant? → **Yes / No** **Algorithms:** Logistic Regression, Decision Trees, SVM, K-NN, Random Forest, Naive Bayes	Predicts a **continuous numerical value**. • Output: any number on a scale • Example: What is the house price? → **$320,000** • Example: Tomorrow's temperature? → **28.5°C** • Example: Stock closing price? → **$145.72** **Algorithms:** Linear Regression, Polynomial Regression, Ridge/Lasso, SVR, Gradient Boosting

■ Section 2: Mathematical Foundations

Essential formulas with plain-English interpretations

2.1 Linear Regression

Linear Regression is the simplest supervised learning algorithm. It finds the best straight line through data points to predict a continuous output. The model learns the values of **m** (slope) and **b** (intercept).

$$y = m \cdot x + b$$

y = predicted output | x = input feature | m = slope (weight) | b = y-intercept (bias)

Multiple Features (General Form)

$$■ = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

Each feature x■ has a corresponding weight w■ learned during training

Intuition: Imagine predicting house price. If size (x■) increases by 1 sq ft, the price goes up by w■ dollars. Each weight tells you how important that feature is.

2.2 Logistic Regression (Sigmoid Function)

Despite its name, **Logistic Regression** is used for **classification**. It takes any input and squashes it into a probability between 0 and 1 using the **sigmoid function**. If the output is ≥ 0.5, the model predicts Class 1; otherwise Class 0.

$$\sigma(z) = 1 / (1 + e^{-z})$$

z = linear combination of features (w·x + b) | σ(z) = probability of Class 1 | e = Euler's number ≈ 2.718

Intuition: If z is very large (strong positive evidence), σ(z) → 1. If z is very negative, σ(z) → 0. At z = 0, the model is 50% uncertain → this is the decision boundary.

2.3 Loss Function – Mean Squared Error (MSE)

A **loss function** measures how wrong the model's predictions are. The model's goal during training is to minimise this value. **MSE** is standard for regression problems.

$$MSE = (1/n) \cdot \Sigma (y■ - ■■)^2$$

n = number of training examples | y■ = actual value | ■■ = predicted value | Σ = sum over all examples

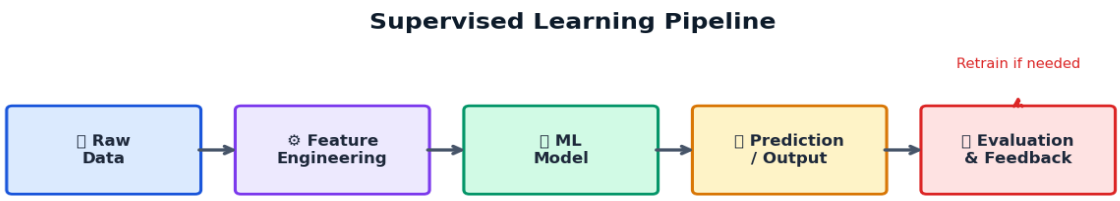
Intuition: Each prediction error is squared (large errors punished more), then averaged. A perfect model has MSE = 0. Training adjusts m and b to minimise this using Gradient Descent.

■ *Cross-Entropy Loss is used for classification (logistic regression) instead of MSE, as it better handles probability outputs and provides stronger gradients.*

■ Section 3: Visual Diagrams & Graphs

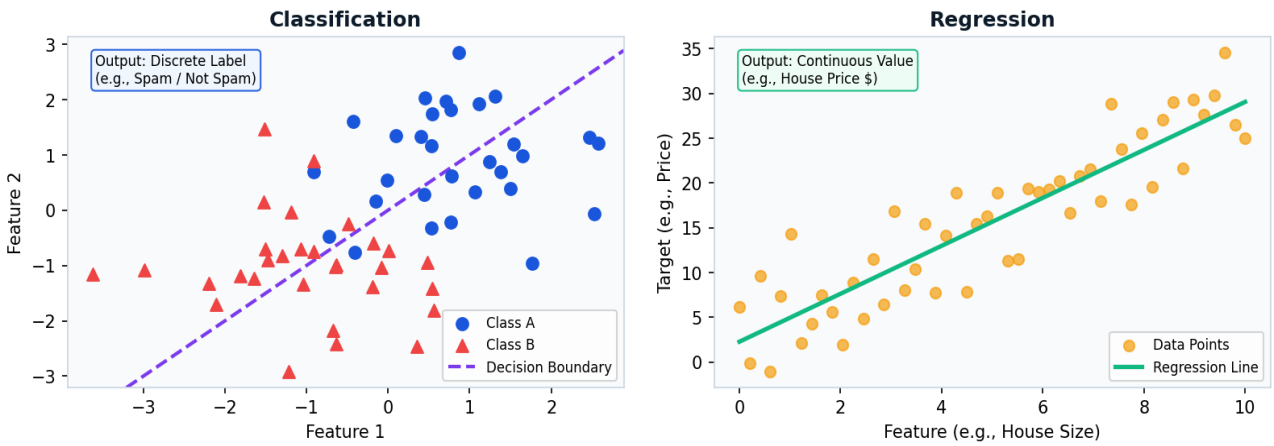
Clean labeled visualisations of key supervised learning concepts

3.1 Supervised Learning Pipeline



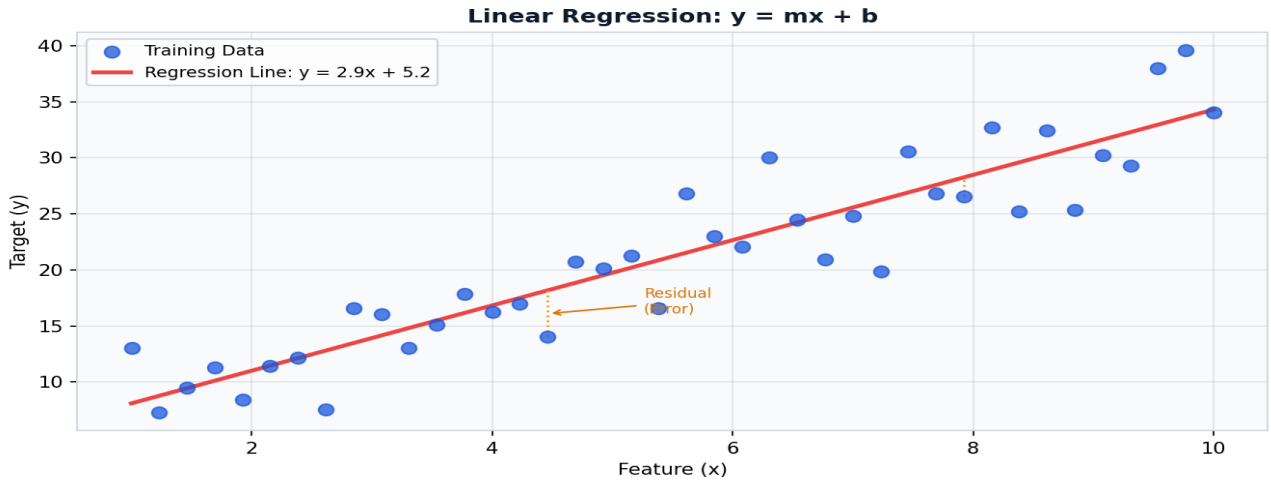
The pipeline shows how raw data flows from collection through feature engineering, into a model, produces outputs, and is evaluated. Poor evaluation triggers retraining.

3.2 Classification vs. Regression (Data View)



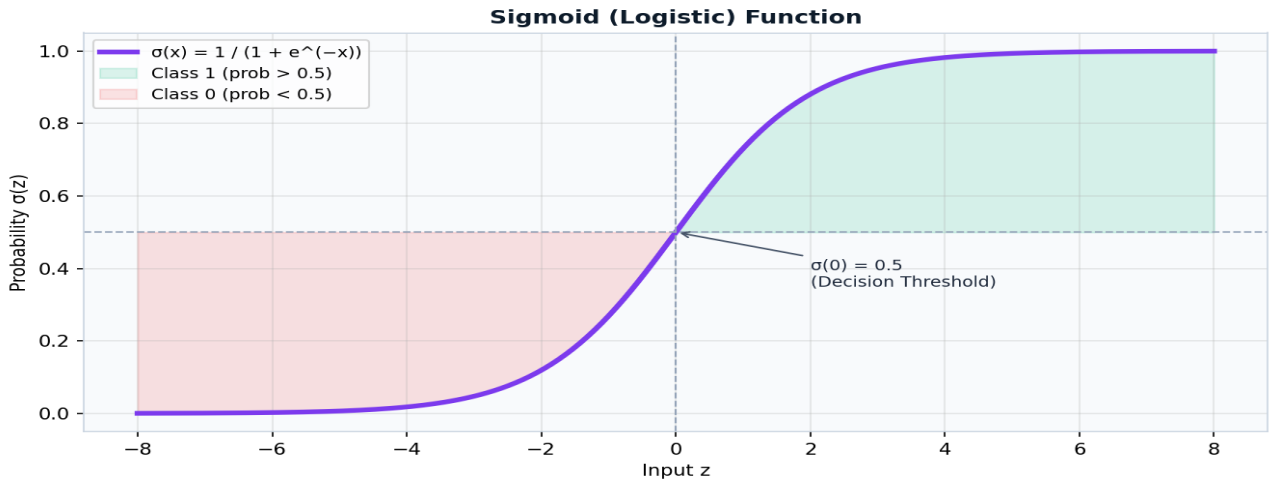
Left: Classification separates data into discrete groups using a decision boundary. Right: Regression fits a continuous line through data to predict numerical values.

3.3 Linear Regression Line



The red line is the best-fit regression line. Yellow dotted lines show residuals (errors). The goal of training is to minimise the total residual (MSE).

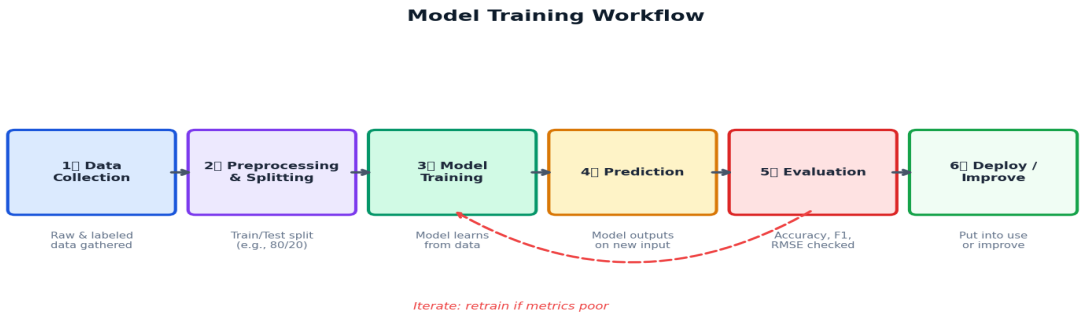
3.4 Sigmoid (Logistic) Curve



The sigmoid curve maps any input to a value between 0 and 1. The green region ($\sigma > 0.5$) represents Class 1 predictions; the red region ($\sigma < 0.5$) represents Class 0 predictions.

■■ Section 4: Model Training Workflow

Step-by-step guide to building a supervised learning model



Step 1 – Data Collection

Gather a large, diverse dataset of **labeled examples**. Quality > Quantity. Biased or incomplete data leads to biased models. Sources: databases, APIs, surveys, sensors, web scraping.

Step 2 – Preprocessing & Train/Test Split

Clean the data: handle missing values, remove duplicates, normalize features. Then split into **Training Set** (~80%) used to learn and **Test Set** (~20%) kept hidden to evaluate real-world performance. A **Validation Set** (from training) is used to tune hyperparameters.

Step 3 – Model Training

Feed training examples to the algorithm. The model adjusts internal parameters (weights) using **Gradient Descent** – repeatedly tweaking values to reduce the loss function. This iterates for many **epochs** until loss converges.

Step 4 – Prediction

Apply the trained model to **new, unseen inputs**. The model uses learned weights to compute outputs. At this stage, **no labels are needed** – inference is purely computational.

Step 5 – Evaluation

Measure performance on the **test set** using appropriate metrics (see Section 5). If metrics are poor → identify the issue (overfit/underfit) and retrain with adjustments.

Step 6 – Deployment & Improvement

A satisfactory model is deployed in production. Continuously monitor for **data drift** (changes in real-world data distribution). Periodically retrain with fresh data to maintain accuracy.

■ Section 5: Evaluation Metrics

Measuring how well your model actually performs

5.1 Classification Metrics

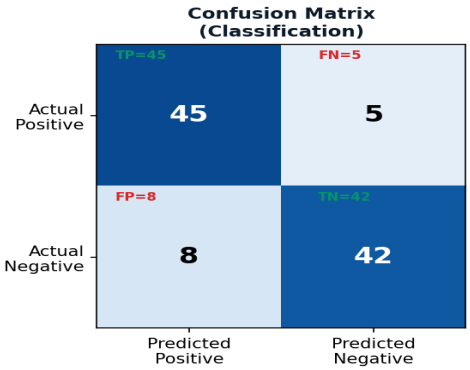
Classification metrics are calculated from the **confusion matrix**: a table of TP (True Positives), FP (False Positives), TN (True Negatives), FN (False Negatives).

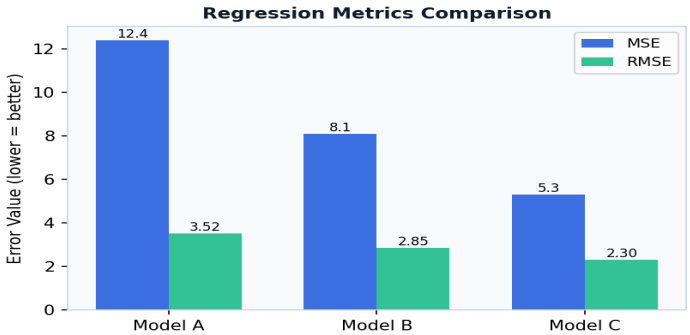
Metric	Formula	Simple Meaning	Best When
Accuracy	(TP + TN) / Total	% of all predictions that are correct	Classes are balanced
Precision	TP / (TP + FP)	Of all predicted positives, how many are real?	False positives are costly (spam filter)
Recall (Sensitivity)	TP / (TP + FN)	Of all real positives, how many did we catch?	Missing positives is costly (disease detection)
F1 Score	2 × (P × R) / (P + R)	Harmonic mean of Precision and Recall	Imbalanced classes
ROC-AUC	Area under ROC curve	Model's ability to distinguish classes	Comparing multiple models

Example: Email spam detector (100 emails: 60 spam, 40 not-spam). Model predicts: 50 spam (45 correct, 5 wrong), 50 not-spam (42 correct, 8 wrong).
→ Accuracy = (45+42)/100 = 87% | Precision = 45/50 = 90% | Recall = 45/60 = 75%

5.2 Regression Metrics

Metric	Formula	Units	Characteristic
MSE (Mean Squared Error)	$(1/n)\ \Sigma(y - ■)^2$	Squared units (e.g., $\$^2$)	Penalises large errors heavily; always ≥ 0
RMSE (Root MSE)	√MSE	Same as output (e.g., $)	Most interpretable; preferred for reporting
MAE (Mean Absolute Error)	$(1/n)\ \Sigma\lvert y - ■\rvert$	Same as output (e.g., $)	Robust to outliers; easier to understand
R^2 Score	1 – (SSres / SStot)	Unitless (0–1)	% of variance explained; 1.0 = perfect


Confusion Matrix
(Classification)
TP=45
FN=5
FP=8
TN=42
45
5
8
42
Actual
Positive
Actual
Negative
Predicted
Positive
Predicted
Negative


Regression Metrics Comparison
MSE
RMSE
Error Value (lower = better)
0
2
4
6
8
10
12
12.4
3.52
8.1
2.85
5.3
2.30
Model A
Model B
Model C

■ Section 6: Real-World Applications

Supervised learning in industry and everyday life

■ Healthcare

• **Disease Diagnosis:** Model trained on patient records (features: age, symptoms, lab results; label: disease present/absent) predicts cancer, diabetes, COVID-19.

• **Medical Imaging:** CNN classifies X-rays or MRI scans as normal/abnormal — used in radiology for tumour detection (classification).

• **Hospital Readmission:** Predicts whether a patient will be readmitted within 30 days using patient history (regression on risk score).

■ Finance

• **Credit Scoring:** Banks predict whether a loan applicant will default (classification: default / no-default) based on income, debt, history.

• **Fraud Detection:** Real-time classification of transactions as fraudulent or legitimate using transaction amount, location, frequency.

• **Stock Price Prediction:** Regression model predicts future stock prices using historical data, volume, market indicators.

■ Email Systems

• **Spam Detection:** Classifies incoming emails as 'spam' or 'not spam' using word frequency, sender reputation, link count as features.

• **Email Categorisation:** Gmail's auto-sort into Primary, Social, Promotions is a multi-class classification task.

• **Phishing Detection:** Identifies malicious emails attempting to steal credentials — critical for cybersecurity.

■ Real Estate

• **House Price Prediction:** Regression model estimates property value using size, location, no. of rooms, age, nearby amenities.

• **Rent Classification:** Classifies properties into price brackets (low/mid/high) for real estate portals.

• **Market Trend Analysis:** Predicts future price trends in a neighbourhood based on economic indicators and historical sales data.

■ Section 7: Quiz – Self Assessment

15 questions across beginner to intermediate difficulty | Do not look at answers first!

Part A: Multiple Choice Questions (MCQ)

Q1. What is the defining characteristic of supervised learning?

A) The model learns without any data
B) The training data includes input-output pairs with labels
C) The model requires no evaluation after training
D) It only works with image data

Q2. Which of the following is a classification problem?

A) Predicting the temperature tomorrow
B) Estimating the salary of an employee
C) Identifying whether an email is spam or not
D) Forecasting quarterly sales revenue

Q3. What does the sigmoid function output?

A) Any real number from $-\infty$ to $+\infty$
B) An integer between 0 and 10
C) A probability value between 0 and 1
D) The class with the highest frequency

Q4. In the equation y = mx + b, what does 'b' represent?

A) The slope of the line
B) The output label
C) The y-intercept (bias term)
D) The input feature

Q5. Which metric is most appropriate when false negatives are very costly?

A) Accuracy
B) Precision
C) Recall (Sensitivity)
D) Mean Squared Error

Q6. A model that performs very well on training data but poorly on test data is:

A) Underfitting
B) Well-generalised
C) Overfitting
D) Perfectly calibrated

Q7. Which loss function is standard for regression problems?

A) Cross-Entropy Loss
B) Mean Squared Error (MSE)
C) Gini Impurity

D) Hinge Loss

Q8. What is the typical train/test split ratio in supervised learning?

A) 50% / 50%
B) 90% / 10%
C) 80% / 20%
D) 60% / 40%

Part B: Short Answer Questions

Q9. Define 'features' and 'labels' in your own words. Give one example of each.

Q10. Explain the difference between classification and regression. Give a real-world example of each.

Q11. What is a loss function and why is it important during model training?

Q12. Why is it important to have a separate test set that the model has never seen?

Q13. You train a spam classifier that achieves 99% accuracy on a dataset where 99% of emails are not spam. Is this model actually useful? Explain why or why not.

Part C: Concept-Based Questions

Q14. Compare Precision and Recall. When would you prioritise Recall over Precision?

Q15. A linear regression model has an RMSE of 50,000 on house price predictions (in $). Another has an RMSE of 30,000. Which is better and why? What other factors might matter?

■ Section 8: Answer Key

Check your answers after completing the quiz

Part A: MCQ Answers

Q	Answer	Explanation
Q1	**B**	Supervised learning is defined by using labeled training data — pairs of inputs and correct outputs.
Q2	**C**	Spam detection outputs a category (spam / not spam), making it classification. The others involve predicting numerical values.
Q3	**C**	The sigmoid function σ(z) = 1/(1+e■■) always outputs a value strictly between 0 and 1, interpretable as a probability.
Q4	**C**	In y = mx + b, 'b' is the y-intercept — the predicted value when x = 0. It's the model's bias term.
Q5	**C**	Recall measures how many actual positives are correctly detected. In cancer diagnosis, missing a real case (FN) is more dangerous than a false alarm.
Q6	**C**	Overfitting means the model memorises training data (including noise) and fails to generalise to new data.
Q7	**B**	MSE = (1/n)Σ(y–■)² is standard for regression. Cross-Entropy is used for classification.
Q8	**C**	The 80/20 split is the most common default, though this varies with dataset size and domain.

Part B & C: Model Answers

Q9:

> Features are the measurable input properties of each data point (e.g., house area, number of rooms). Labels are the correct output values we want the model to predict (e.g., house price = $250,000). Features are what you know; the label is what you want to find out.

Q10:

> Classification predicts a discrete category — e.g., 'Is this tumour malignant? Yes/No.' Regression predicts a continuous number — e.g., 'What will be the tumour growth rate? 2.3 mm/month.' The key difference is the type of output.

Q11:

A loss function quantifies the error between the model's predictions and the actual labels. During training, the optimiser (e.g., Gradient Descent) adjusts model weights to minimise this value. Without a loss function, there is no objective for the model to improve toward.

Q12:

If the model sees test data during training, it can memorise answers rather than learning general patterns. A hidden test set gives an unbiased estimate of real-world performance. This is called avoiding 'data leakage'.

Q13:

No. This is the 'accuracy paradox'. A naive classifier that always predicts 'not spam' would also get 99% accuracy. You need Precision and Recall on the spam class specifically. The model is useless if it never catches any spam.

Q14:

Precision = how many predicted positives are actually positive. Recall = how many actual positives were captured. Prioritise Recall when missing positives is dangerous (cancer diagnosis, fraud detection) — even at the cost of more false alarms. Prioritise Precision when false alarms are costly (spam filters, legal evidence).

Q15:

RMSE of 30,000 is better (lower error). However, context matters: on a $500,000 median price, 30k error is ~6% which may be acceptable, while on a $60,000 property it's 50% which is not. Also consider MAE (robustness to outliers) and R^2 (variance explained) for a full picture.

■ Section 9: Glossary

Quick-reference definitions for all key terms in this chapter

Feature
An individual measurable property or characteristic of the data used as input to the model. Also called an attribute, predictor, or independent variable. Example: a person's age or height.

Label
The output variable / target we want the model to predict. Also called the dependent variable, target, or ground truth. Example: whether an email is spam (1) or not (0).

Model
A mathematical function learned from training data that maps input features to output predictions. Examples: Linear Regression, Decision Tree, Neural Network.

Training Set
The labelled subset of data used to teach the model. The model sees both features and labels during this phase and adjusts its parameters.

Test Set
A held-out portion of labelled data never shown to the model during training. Used to provide an unbiased evaluation of final model performance.

Overfitting
When a model learns the training data too well, including noise and random fluctuations, causing it to perform poorly on new, unseen data. Sign: high training accuracy, low test accuracy.

Underfitting
When a model is too simple to capture the underlying patterns in the data, performing poorly even on training data. Sign: low accuracy on both training and test sets.

Regression
A type of supervised learning task where the output is a continuous numerical value. Examples: predicting house price, temperature, stock value.

Classification
A supervised learning task where the output is a discrete class label. Examples: spam/not-spam, cat/dog/bird, malignant/benign.

Loss Function
A function that measures the difference between predicted and actual values. The training objective is to minimise this. Examples: MSE (regression), Cross-Entropy (classification).

Gradient Descent
An optimisation algorithm that iteratively adjusts model parameters in the direction that most reduces the loss function, like descending a hill to find the lowest point.

Hyperparameter
A configuration value set before training that controls the learning process (e.g., learning rate, number of trees, max depth). Not learned from data — must be tuned manually or via search.

Bias-Variance Tradeoff

The tension between a model being too simple (high bias → underfitting) and too complex (high variance → overfitting). The goal is to find the sweet spot that generalises well.

Cross-Validation

A technique where the dataset is split into k folds; the model is trained k times, each time using a different fold as validation. Provides a more robust estimate of model performance.

■ *Study Tip: Try to write each glossary definition in your own words without looking. This active recall technique is one of the most effective revision strategies.*

End of Chapter 1 – Supervised Learning

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